

Lecture #11: Solving the Linear Programming Problem

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We know that the objective function of any linear programming problem could be written as

$$\mathbf{c}^\top \mathbf{x}.$$

This has to equal something, so we say that $\mathbf{c}^\top \mathbf{x} = k$. Then the goal is to optimize this hyperplane. We can think of this as finding the first set of feasible solutions that intersect the solution hyperplane.



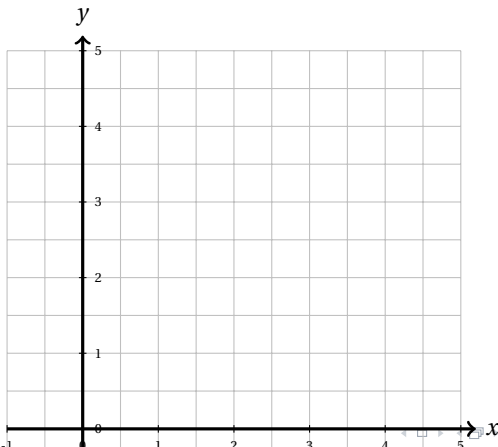
Consider the linear programming problem

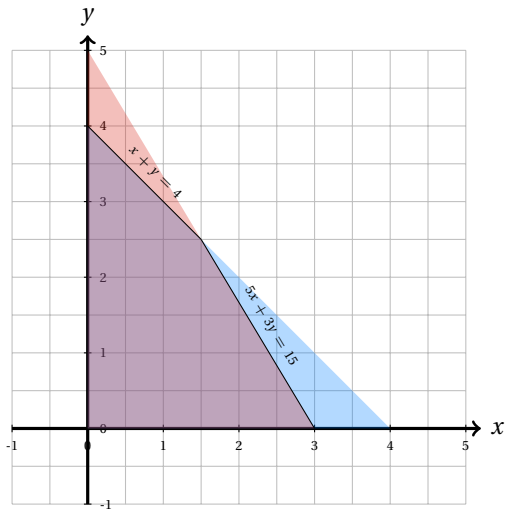
maximize $z = 4x + 3y$ subject to

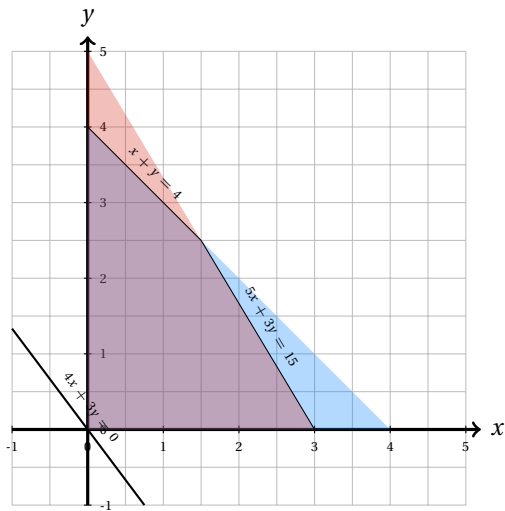
$$x + y < 4$$

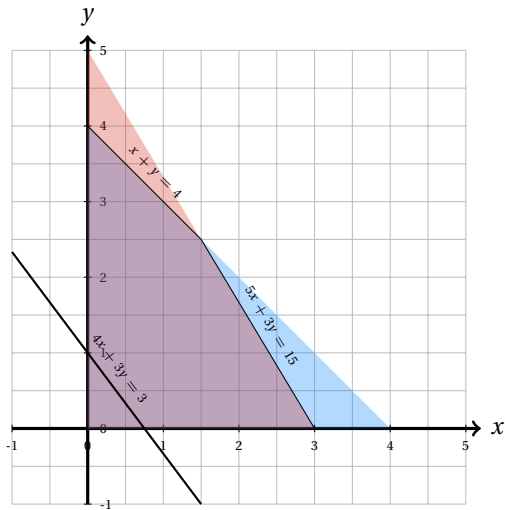
$$5x + 3y < 15$$

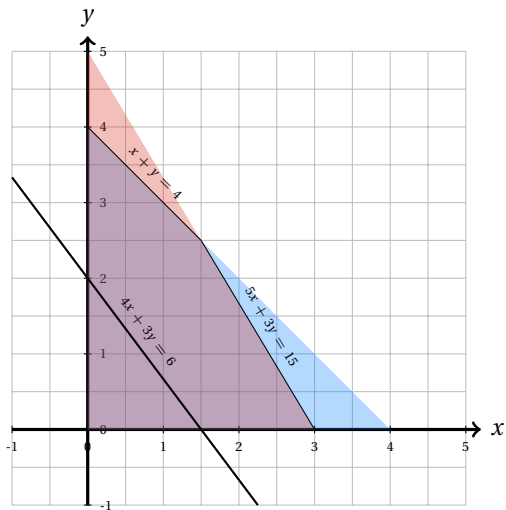
$$x, y > 0$$

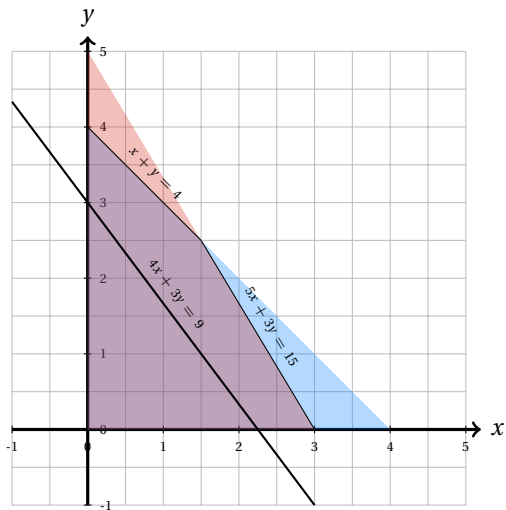


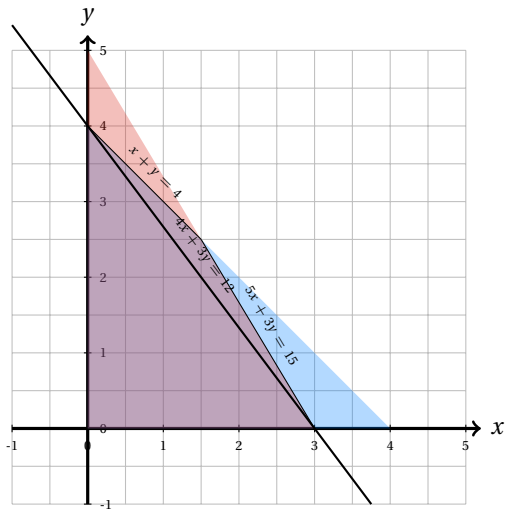


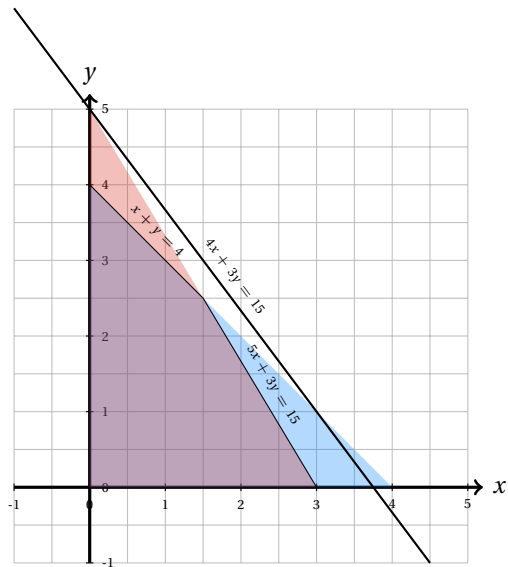


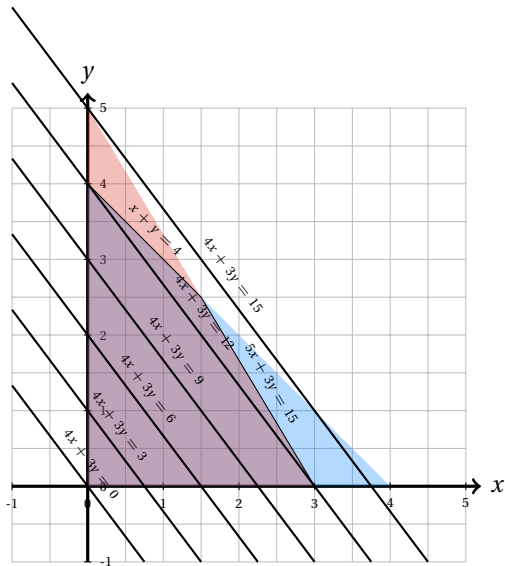




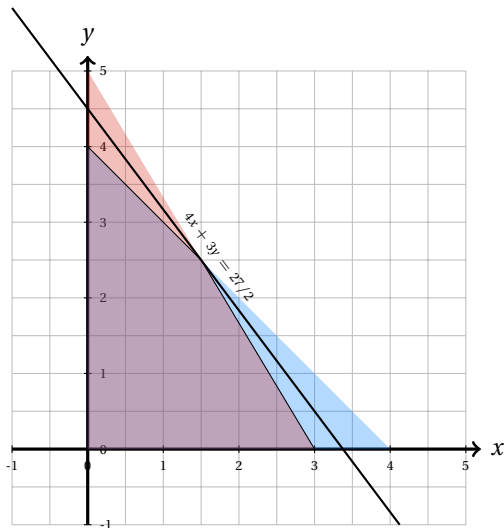








We are able to see that the solution is somewhere between $z = 12$ and $z = 15$. In fact, the solution is when $z = 27/2$, leading to the solution of $(x, y) = (3/2, 5/2)$.



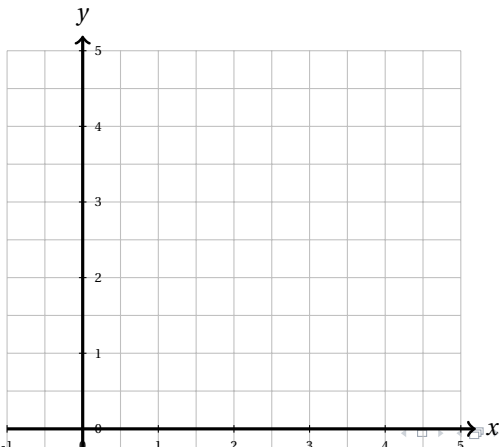
Consider the linear programming problem

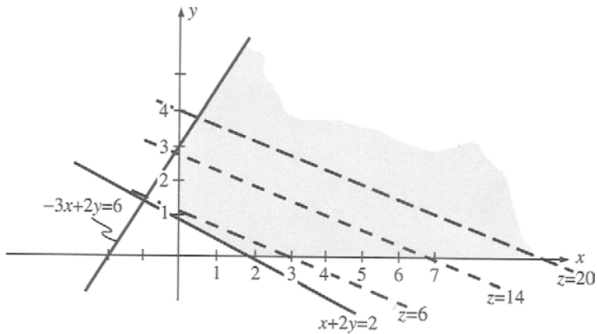
Maximize $z = 2x + 5y$ subject to

$$-3x + 2y < 6$$

$$x + 2y > 2$$

$$x > 0, y > 0$$





We first show that if $\mathbf{x}_1$ and $\mathbf{x}_2$ are two feasible solutions, then any point on the line segment joining these two points is also a feasible solution. The line segment joining $\mathbf{x}_1$ and $\mathbf{x}_2$ is defined as

$$\{\mathbf{x} \in \mathbb{R}^n | \mathbf{x} = p\mathbf{x}_1 + (1 - p)\mathbf{x}_2\}, \quad 0 \leq p \leq 1$$

We define the *end points* as $\mathbf{x}_1, \mathbf{x}_2$, and any points in between them are *interior points*



Now consider the objective function for two feasible points where both
 $\mathbf{c}^\top \mathbf{x}_1 = \mathbf{c}^\top \mathbf{x}_2 = k$?



Now lets $\mathbf{c}^\top \mathbf{x}_1 \leq \mathbf{c}^\top \mathbf{x}_2$, then we have

Definition

A set C is convex if for all $x, y \in C$ and $p \in [0, 1]$,

$$px + (1 - p)y \in C.$$



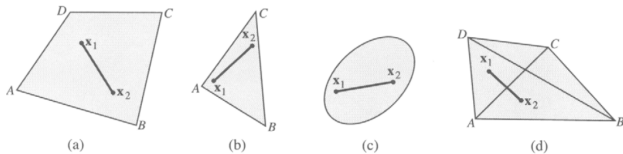


FIGURE 1.9

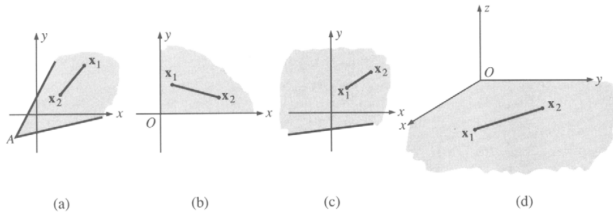


FIGURE 1.10

